Training-Free Group Relative Policy Optimization

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Background: The Rise of LLM Agents

- **General Capabilities:** LLMs excel in problem-solving, research, and code generation.
- **Specialized Limitations:**
  - Struggle in real-world domains requiring specific APIs/tools.
  - Out-of-the-box performance is often suboptimal.
- **Current Solution:** Agentic Reinforcement Learning (Agentic RL) via parameter tuning (e.g., GRPO).
- **Challenges in Parameter Tuning:** Computational Cost, Poor Generalization, Data Scarcity, Diminishing Returns.

The Proposed Method: Training-Free GRPO

Core Concept: Shift policy optimization from *Parametric Space* to *Contextual Space*.

How it Works

1. **Rollout Generation:** Generate a group of diverse outputs for each query.
2. **Self-Introspection:** LLM introspects on rollouts to distill a Semantic Advantage.
3. **Iterative Refinement:** Evolving experiential knowledge acts as token priors without gradient updates.

Result: Significantly enhances frozen models (e.g., DeepSeek-V3.1) with only dozens of samples.

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Recap: Vanilla GRPO Mechanism

- **Rollout:** Policy π_θ generates G outputs $\{o_1, \ldots, o_G\}$ for query q.
- **Reward:** Reward model R assigns scalar scores r_i.
- **Numerical Advantage:** Relative quality is quantified as:

$$\hat{A}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r)}$$

- **Optimization:** Maximizes $J_{GRPO}(\theta)$ via gradient ascent to update parameters θ.

Methodology: Training-Free GRPO

Translating group-based evaluation into a **non-parametric** process. Instead of updating the parameters θ, they leave θ permanently frozen and maintain an external experiential knowledge E.

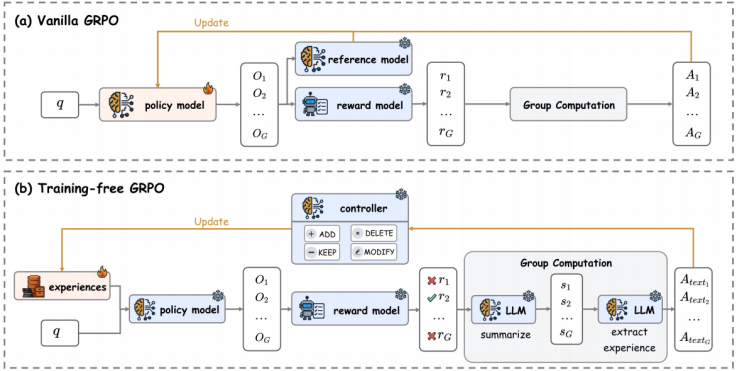

Figure 2. Comparison of vanilla GRPO and Training-Free GRPO.

Mechanism Breakdown (1): Rollout & Conditioned Policy

- **In-Context Optimization:** The policy $\pi_\theta(o_i|q, E)$ treats the experience library E as a dynamic prompt prefix.
- **Group Generation:** For each query, G independent rollouts are generated to explore the policy space.
- **Reward Grounding:** Each rollout is evaluated by a Reward Model $R(q, o_i)$, providing the necessary "ground truth" for relative comparison.

Mechanism Breakdown (2): Distilling Semantic Advantage

The core innovation: Converting numerical gaps into logical insights.

The "Introspection" Pipeline

- **Summary (s_i):** LLM condenses complex outputs into structured summaries to reduce noise.
- **Experience Extraction (A_{text}):** The LLM analyzes the *relative success/failure* of different summaries within the group.
- **The Result:** A concise natural language insight that explains *why* certain actions led to higher rewards.

Key Insight: A_{text} encodes the gradient direction in natural language tokens.

Mechanism Breakdown (2): Distilling Semantic Advantage

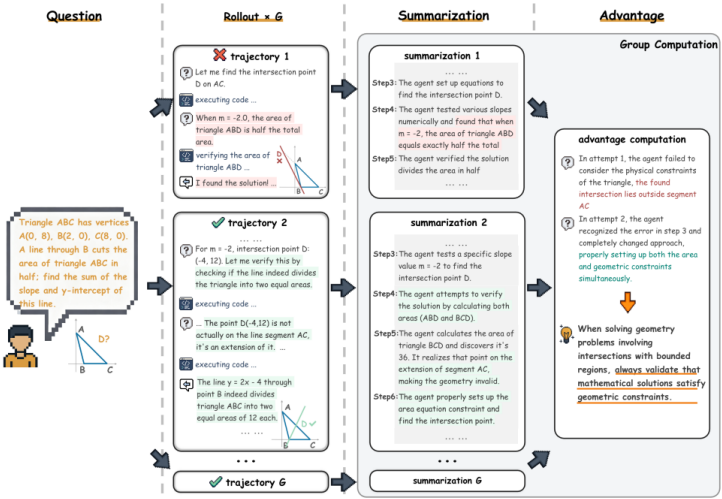

Figure 3. Example of a Training-Free GRPO learning step.

Mechanism Breakdown (3): Non-Parametric Optimization

Instead of adjusting weights w_{ij}, they perform **Semantic Updates** on E:

Add	Append novel successful strategies derived from A_{text}.
Delete	Prune harmful or redundant experiences that caused low rewards.
Modify	Refine existing rules based on evolving contextual evidence.
Keep	The experience library E remains unchanged.

- **Stability:** The frozen base model acts as a powerful anchor (similar to the KL-constraint in PPO/GRPO).
- **Efficiency:** Achieves policy shift in a few "inference steps" rather than millions of "backpropagation steps."

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Mathematical Reasoning

- **Benchmarks:** Evaluation on **AIME 2024** and **AIME 2025** (Complex, out-of-domain mathematical challenges).
- **Metric: Mean@32** (Average Pass@1 score from 32 independent runs for statistical reliability).
- **Base Model:** DeepSeek-V3.1-Terminus.
- **Configurations:**
  - **Direct Prompting:** Text-only input/output.
  - **ReAct:** Tool-augmented reasoning using a Code Interpreter (CI).
- **Learning Phase:** 100 random samples from DAPO-Math-17K; 3 epochs; Group Size $= 5$.

Main Results: Performance Gains

- Training-Free GRPO achieves substantial absolute gains (+2.7% on AIME24 and +5.4% on AIME25) using only **100 examples** and **zero gradient updates**.
- It compares Training-free GRPO on DeepSeek-V3.1 against RL baselines on Qwen2.5-32B. This is an unfair setup. The performance gap likely stems from the model's inherent strength rather than the method itself.

Table 3. Mean@32 with smaller LLMs on AIME 2024 and AIME 2025 benchmarks (%).

Method	Learning Cost	Model	Tool	AIME24	AIME25
ReAct [1]	-	*Qwen2.5-32B-Instruct*	CI	29.6	23.1
ZeroTIR [5]	$\approx$ \$20,000	*Qwen2.5-32B-Instruct*	CI	56.7	33.3
SimpleTIR [6]	$\approx$ \$20,000	*Qwen2.5-32B-Instruct*	CI	59.9	49.2
ReTool [4]	$\approx$ \$10,000	*Qwen2.5-32B-Instruct*	CI	67.0	49.3
AFM [16]	$\approx$ \$10,000	*Qwen2.5-32B-Instruct*	CI	66.7	59.8
ReAct [1]	-	*Qwen3-32B (Non-Thinking)*	CI	29.1	19.5
+ Training-Free GRPO	$\approx$ \$4	*Qwen3-32B (Non-Thinking)*	CI	33.5 (↑4.4)	25.4 (↑5.9)
ReAct [1]	-	*Qwen2.5-72B-Instruct*	CI	13.5	9.6
+ Training-Free GRPO	$\approx$ \$3	*Qwen2.5-72B-Instruct*	CI	14.9 (↑1.4)	11.4 (↑1.8)
ReAct [1]	-	*DeepSeek-V3.1-Terminus*	CI	80.0	67.9
+ Training-Free GRPO	$\approx$ \$18	*DeepSeek-V3.1-Terminus*	CI	82.7 (↑2.7)	73.3 (↑ 5.4)

Learning Dynamics & Efficiency

Key Insights:

- **Step-wise Improvement:** Performance on the training set and out-of-domain benchmarks improves steadily over 3 steps.
- **Action Efficiency:** The average number of tool calls **decreases**, suggesting the agent learns to avoid redundant or erroneous actions.
- **Shortcuts:** Semantic advantage optimization helps discover more efficient reasoning paths.

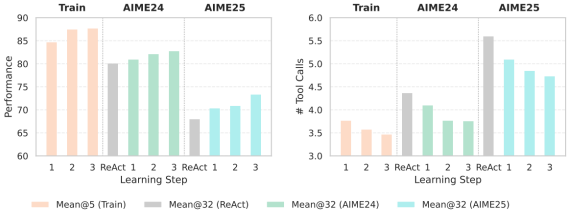

Figure 4. Statistics at each Training-Free GRPO step with tool use and DeepSeek-V3.1-Terminus.

Robustness & Ablation Analysis

Table 2. Ablation study on DeepSeek-V3.1-Terminus (Mean@32, %).

Method	Training Set	AIME24	AIME25
ReAct [1]	-	80.0	67.9
ReAct [1] + Directly Generated Experiences	-	79.8	67.3
ReAct [1] + Training-Free GRPO (w/o ground truths)	DAPO-100	80.7	68.9
ReAct [1] + Training-Free GRPO (w/o group computation)	DAPO-100	80.4	69.3
ReAct [1] + Training-Free GRPO	DAPO-100	**82.7**	**73.3**

- **Effectiveness of Learned Experience:** Directly generated experiences (without GRPO optimization) fail to improve performance, validating the specific optimization method.
- **Reward Signal Robustness:** Without ground truth answers, the model uses implicit majority voting and self-reflection. It still achieves **80.7%** on AIME24, proving utility in data-scarce domains.
- **Group Computation:** Reducing Group Size to 1 significantly harms performance, confirming that **relative trajectory comparison** is essential for semantic advantage.

Web Searching: Agents in Realistic Environments

- **Dataset:** 100 samples from **AFM** (RL-collected web interactions) for learning; **WebWalkerQA** for evaluation.
- **Task Complexity:** Requires understanding both natural language and complex DOM/web structures.
- **Performance Boost:** DeepSeek-V3.1-Terminus Pass@1 increased from **63.2%** $\rightarrow$ **67.8%**.

Table 4. Pass@1 on WebWalkerQA (%).

Method	Training Set	Model	pass@1
ReAct [1]	-	*DeepSeek-V3.1-Terminus*	63.2
+ Training-Free GRPO	AFM-100	*DeepSeek-V3.1-Terminus*	**67.8** (↑4.6)

Ablation Studies: What Drives the Gains?

- **The Failure of Raw Experience:** Using directly generated experiences (without GRPO optimization) **degraded** performance (64.7% vs. 66.7% ReAct baseline).
- **Robustness to Labels:** Even **without ground truth**, the method maintained baseline Pass@1 while improving **Pass@3 to 78.4%**.
- Experience-based optimization is highly dependent on the underlying model's inherent reasoning and tool-use capabilities.

Table 5. Ablation results on WebWalkerQA subset (%).

Method	Training Set	Model	pass@1	pass@3
ReAct [1]	-	*QwQ-32B*	27.5	43.1
ReAct [1] + Training-Free GRPO	AFM-100	*QwQ-32B*	25.5	45.1
ReAct [1]	-	*DeepSeek-V3.1-Terminus*	66.7	74.5
ReAct [1] + Directly Generated Experiences	AFM-100	*DeepSeek-V3.1-Terminus*	64.7	76.5
ReAct [1] + Training-Free GRPO (w/o ground truths)	AFM-100	*DeepSeek-V3.1-Terminus*	66.7	78.4
ReAct [1] + Training-Free GRPO	AFM-100	*DeepSeek-V3.1-Terminus*	**68.6**	**78.4**

Cross-Domain Robustness: Generalist vs. Specialist

- **The Specialization Trade-off:** Parameter-tuned models (e.g., ReTool) excel in their training domain but collapse in others.
- **Comparison:**
  - **Math-tuned ReTool:** Drops to **18.3%** on Web tasks (lower than non-tuned ReAct).
  - **Training-Free GRPO:** Maintains State-of-the-Art performance in **both** domains simultaneously by swapping "Learned Experience" modules.
- It seems this is trivial because the paper is simply swapping domain-specific "experience libraries."

Table 6. Cross-domain transferability (Averaged pass@1, %).

Method	Learned Domain	Math Reasoning		Web Searching
		AIME24	AIME25	WebWalker
ReAct [1] *(Qwen2.5-32B-Instruct)*	-	29.6	23.1	31.9
ReTool [4] *(Qwen2.5-32B-Instruct)*	Math	67.0	49.3	18.3
MiroThinker [10] *(Qwen3-32B)*	Web	43.5	36.8	53.6
Training-Free GRPO *(DeepSeek-V3.1-Terminus)*	Math / Web	**82.7**	**73.3**	**67.8**

Economic Benefits: Two Orders of Magnitude Savings

Training Cost Comparison:

- **Vanilla RL (ReTool):** $\approx$ 20,000 GPU hours $\approx$ **$10,000**.
- **Proposed Method:** 100 samples / 3 steps $\approx$ **$18**.

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Final Summary: Training-Free GRPO

Core Conclusion

- **Paradigm Shift:** Successfully shifts policy optimization from **Parameter Space** to **Context Space**, proving that frozen LLMs can be steered via iterative "Experience Distillation."
- **Generalization through Modularity:** Preserves the base model's general capabilities by using "plug-and-play" experience libraries, avoiding the catastrophic forgetting of fine-tuning.

Critical Weaknesses

- **Model Dependency:** Acts as a **"Force Multiplier"** rather than a "Fix"; showing heavy reliance on baseline reasoning.
- **Conceptual Framing:** "Training-Free GRPO" is technically an *oxymoron*; it is closer to **Iterative Prompt Engineering** than true Reinforcement Learning.
- **Library Scalability:** Long-term accumulation of experiences may exceed the **context window**, leading to retrieval overhead or performance degradation.

Takeaway

While Training-Free GRPO is a highly practical and cost-effective tool for frontier models, its scientific rigor and scaling laws for smaller models require further validation.

Thank you!